Carbonly — User Stories & Acceptance Criteria

v2.0 · April 2026

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Version: 2.0 **Last updated:** 2026-04-15 **Status:** Active development **Format:** Agile user stories with INVEST principles · Given / When / Then acceptance criteria

This document captures every user-facing behaviour in the Carbonly platform. Each story follows the standard template (**As a**, **I want**, **So that**), is sized to fit in a single sprint, and is paired with explicit **validation rules**, **business rules**, **acceptance criteria**, and **edge cases**.

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1. Personas

👨‍💼 Adam — Platform Super Admin

- **Role:** `SUPER_ADMIN`
- **Context:** Works at Carbonly, owns the platform taxonomy and tenant lifecycle
- **Goals:** Onboard customers, maintain emission factors, monitor platform revenue
- **Tools used:** `/admin/*` routes, Stripe dashboard, Prisma Studio (ops)

👩‍💼 Emma — Entity Administrator (Customer side)

- **Role:** `ADMIN` within an Entity (optionally `EXPERT` globalRole)
- **Context:** Head of Sustainability at Acme Corp
- **Goals:** Manage team members, configure yearly baseline, approve billing, publish annual reports
- **Tools used:** Dashboard, Reports, Settings (billing section)

👨‍🔬 Carlos — Carbon Expert

- **Role:** `EXPERT`
- **Context:** Environmental consultant embedded at Acme Corp
- **Goals:** Enter monthly activity data, run calculations, verify numbers before reporting
- **Tools used:** Scope 1/2/3 calculators, emissions records table, dashboard

👤 Priya — Visitor / Prospect

- **Role:** Unauthenticated
- **Context:** Visits carbonly.io after seeing an ad
- **Goals:** Understand what the product does and whether to sign up
- **Tools used:** Landing page, pricing section, register flow

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2. Story conventions

Each story has the following structure:

- **ID** — unique identifier (e.g. `US-001`)
- **Title** — short imperative description
- **As a** / **I want** / **So that** — classic user-story framing
- **Persona** — primary actor
- **Priority** — Must / Should / Could / Won't (MoSCoW)
- **Story points** — Fibonacci estimate (1, 2, 3, 5, 8, 13)
- **Dependencies** — other stories that must ship first
- **Validations** — input rules enforced by the UI, API, and/or database
- **Business rules** — non-validation rules driven by domain requirements
- **Acceptance criteria** — Given / When / Then scenarios that must pass
- **Edge cases** — unusual paths that must be handled

Priority legend

PRIORITY	MEANING
Must	Required for MVP release
Should	Important but can ship in a follow-up
Could	Nice-to-have, ships when time allows
Won't	Explicitly deferred

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3. Epic 1 — Authentication & Account Management

Goal: Allow users to securely access the platform and manage their credentials.

US-001 · Register a new account

As a visitor **I want** to create a Carbonly account with my name, email, and password **So that** I can sign in and use the platform

- **Persona:** Priya
- **Priority:** Must
- **Story points:** 3

- **Dependencies:** None

Validations

FIELD	RULE
Full name	Required · 2–80 chars · letters, spaces, hyphens, apostrophes
Email	Required · valid RFC 5322 format · unique across `User.email`
Password	Required · min 8 chars · must contain at least one letter and one digit
Confirm password	Must equal `password` field

Business rules

- Password is stored as a **bcryptjs hash** (cost factor 10). Plain text is never persisted.
- The email comparison is **case-insensitive** (lowercased on both write and read).
- On successful registration, no session is created — the user is redirected to `/login` to sign in explicitly.
- Duplicate-email attempts return `409 Conflict` with error `"Email already registered"`.

Acceptance criteria

1. **Given** I am on `/register` **When** I submit valid name, email, and password **Then** a `User` row is created, I see a success message, and I'm redirected to `/login?registered=1`.
2. **Given** the email `existing@example.com` already exists **When** I submit the form with that email **Then** I see the error "Email already registered" and no new user is created.
3. **Given** my password is `abc` **When** I submit the form **Then** I see "Password must be at least 8 characters".
4. **Given** I have JavaScript disabled **When** I submit the form **Then** server-side validation still runs and the same errors surface.

Edge cases

- Whitespace-only name is rejected.
- Trailing whitespace on email is trimmed before saving.
- Unicode names (e.g. `Ayşe Yılmaz`) are accepted.
- Passwords longer than 72 bytes are accepted but bcrypt only hashes the first 72 bytes (documented limitation).

US-002 · Sign in with email and password

As a registered user **I want** to sign in with my email and password **So that** I can access my organisation's data

- **Persona:** Carlos
- **Priority:** Must
- **Story points:** 3
- **Dependencies:** US-001

Validations

FIELD	RULE
Email	Required · valid format
Password	Required · 1+ chars

Business rules

- Authentication handled by **NextAuth v5 CredentialsProvider**.
- Password comparison uses `bcrypt.compare()`.
- Failed attempts return a **generic error** (`"Invalid email or password"`) to prevent enumeration.
- Successful sign-in issues a **JWT session cookie** (HttpOnly, SameSite=Lax, signed with `NEXTAUTH_SECRET`).
- JWT contains `{ sub, globalRole, entityId, entitySlug, entityRole }`.

Acceptance criteria

1. **Given** valid credentials **When** I submit the login form **Then** a session cookie is set and I'm redirected to `/dashboard`.
2. **Given** a wrong password **When** I submit **Then** I see "Invalid email or password" and no cookie is set.
3. **Given** an email that doesn't exist **When** I submit **Then** I see the same generic error (no account enumeration).
4. **Given** I'm already signed in **When** I visit `/login` **Then** I'm redirected to `/dashboard` (no re-login prompt).

US-003 · Sign in with Google

As a registered user **I want** to sign in with my Google account **So that** I don't need to remember another password

- **Persona:** Emma
- **Priority:** Should
- **Story points:** 2
- **Dependencies:** US-001, `GOOGLE_CLIENT_ID` + `GOOGLE_CLIENT_SECRET` configured

Business rules

- Google OAuth uses the **Authorization Code flow** via NextAuth's `GoogleProvider`.
- On first sign-in, a `User` + `Account` row is created by `PrismaAdapter`.
- The user's Google profile name and image are imported; password remains null.

Acceptance criteria

1. **Given** I click "Sign in with Google" **When** I complete Google's consent screen **Then** a `User` is created or matched by email, and I'm redirected to `/dashboard`.
2. **Given** my email is already registered via credentials **When** I sign in with Google using the same email **Then** a new `Account` row links Google to my existing user (no duplicate `User`).

US-004 · Sign out

As a signed-in user **I want** to sign out **So that** my session is terminated on shared devices

- **Persona:** Carlos
- **Priority:** Must
- **Story points:** 1

Acceptance criteria

1. **Given** I'm signed in **When** I click "Sign out" in the topbar **Then** the session cookie is cleared and I'm redirected to `/login`.
2. **Given** I've signed out **When** I attempt to access `/dashboard` via the back button **Then** middleware redirects me to `/login`.

US-005 · Protected routes enforce session

As the system **I want** to block unauthenticated access to tenant pages **So that** data isolation is guaranteed

- **Persona:** (System behaviour)
- **Priority:** Must
- **Story points:** 2

Business rules

- **Middleware** (`src/middleware.ts`) runs at the edge on every request except `/api/auth/*`, `/api/webhooks/*`, `/login`, `/register`, and public assets.
- Middleware uses `authConfig` (lightweight, no Prisma) to check JWT presence.
- Unauthenticated requests are redirected to `/login`.
- Tenant pages (`/dashboard`, `/scope/*`, `/emissions`, `/reports`, `/settings`) and admin pages (`/admin/*`) both require a valid session.

Acceptance criteria

1. **Given** I have no session cookie **When** I request `GET /dashboard` **Then** I get a 307 redirect to `/login`.
2. **Given** I have a valid session but am not a super admin **When** I request `/admin` **Then** in demo mode I am allowed through (production: redirect to `/dashboard`).
3. **Given** any authenticated request **When** the server queries tenant data **Then** the query filters by `entityId` from `session.user.entityId` — never trusting client-sent IDs.

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4. Epic 2 — Onboarding & Workspace Navigation

Goal: Help users find their way around the app after signing in.

US-010 · Land on the dashboard after sign-in

As Carlos **I want** to see my emissions dashboard as my landing page **So that** I get value within seconds of signing in

- **Priority:** Must · **Story points:** 1

Acceptance criteria

1. **Given** I sign in successfully **When** the redirect completes **Then** I see the dashboard for the current year with total emissions and scope cards.

US-011 · Navigate via the persistent sidebar

As any user **I want** a sidebar with links to all major areas **So that** I can move between sections without thinking

- **Priority:** Must · **Story points:** 2

Business rules

- Sidebar is **permanent** on desktop (≥ 992px viewport width) and **temporary** (hamburger-triggered) on mobile.
- Active route is visually highlighted.
- Sidebar shows:
  - **Dashboard** · **All Records** (Emissions)
  - **Calculator** section: Scope 1, Scope 2, Scope 3
  - **Reporting** section: Reports, Settings
  - **Platform** section: Admin Panel (if user has permission or in demo mode)
- The Carbonly logo links back to `/`.
- The footer shows user's name and role.

Acceptance criteria

1. **Given** I'm on the dashboard **When** I click "All Records" **Then** I navigate to `/emissions` and the sidebar highlights that item.
2. **Given** I'm on a mobile viewport (<992px) **When** I tap the hamburger icon **Then** a temporary drawer slides in, and tapping an item closes the drawer after navigation.

US-012 · Switch role (demo mode)

As a demo visitor **I want** to switch between the admin view and the expert view without re-login **So that** I can explore both perspectives quickly

- **Priority:** Could (demo-only) · **Story points:** 2

Business rules

- In the **admin sidebar** footer, a "Switch role" section offers a button labelled **Expert View**.
- Clicking it calls `signIn("credentials", { email: "expert@acme-corp.com", password: "expert1234", redirect: true, callbackUrl: "/dashboard" })`.
- This replaces the current session entirely (NextAuth v5 behaviour).

Acceptance criteria

1. **Given** I'm signed in as super admin on `/admin` **When** I click "Expert View" **Then** my session is replaced by the expert account and I'm redirected to `/dashboard`.
2. **Given** the expert's dashboard sidebar includes an "Admin Panel" link **When** I click it **Then** I'm taken to `/admin` without a redirect block.

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5. Epic 3 — Emissions Dashboard

Goal: Give tenant users instant insight into their carbon footprint.

US-020 · View total emissions hero panel

As Emma **I want** to see a large "Total Emissions" hero card at the top of my dashboard **So that** I get the headline number at a glance

- **Priority:** Must · **Story points:** 3

Business rules

- Hero card shows **total tCO_2e** for the selected year, with 1 decimal place.
- Below the number, a **12-month sparkline** shows monthly distribution.
- Two chips compare current year to:
  - **Prior year** (`yoyChangePct` %, green if lower, red if higher)
  - **Baseline year** (2021) (`vsBaselinePct` %, same colour logic)
- If there is no data for the selected year, the card shows "No data yet — start by logging emissions in the Scope calculators".

Acceptance criteria

1. **Given** 450 emission records totalling 1,248.5 tCO_2e for 2025 **When** I view `/dashboard?year=2025` **Then** the hero shows "1,248.5 t CO_2e" with the correct YoY and baseline chips.
2. **Given** the prior year had 1,420 tCO_2e **When** I'm on 2025 (1,248.5 tCO_2e) **Then** the YoY chip reads "↓ 12%" and is coloured green.
3. **Given** 2025 is the selected year and it's the baseline **When** the card renders **Then** no "vs baseline" chip is shown (would always be 0%).

US-021 · View scope breakdown cards

As Carlos **I want** to see individual cards for Scope 1, Scope 2, and Scope 3 **So that** I can see where my emissions concentrate

- **Priority:** Must · **Story points:** 2

Business rules

- Three cards in a row (or stacked on mobile via Bootstrap `col-12 col-md-4`).
- Each card shows: scope label, sub-label (e.g. "Direct", "Energy", "Value Chain"), tCO_2e value, percentage of total, and a progress bar.
- Cards are **clickable** — clicking expands a drill-down panel below showing top activities for that scope.
- Colour mapping: Scope 1 green, Scope 2 teal/blue, Scope 3 amber.

Acceptance criteria

1. **Given** Scope 1 = 342.1 t, Scope 2 = 518.4 t, Scope 3 = 388 t (total 1248.5) **When** I view the cards **Then** Scope 2 shows "42%" and its bar fills 42% of the track.
2. **Given** I click Scope 2 **When** the panel opens **Then** I see Scope 2 activities sorted by CO_2e descending.
3. **Given** I click Scope 2 again **When** the same card is clicked **Then** the panel collapses.

US-022 · View scope breakdown pie chart

As Emma **I want** an interactive pie chart showing the scope split **So that** I have a visual proportion at a glance

- **Priority:** Should · **Story points:** 3

Business rules

- Recharts `PieChart` with 3 slices.
- Hovering a slice shows tooltip with tCO_2e value.
- Clicking a slice opens a drill-down sub-view showing the top activities in that scope.
- Drill-down includes a "← Back" button to return to the overall view.

US-023 · View monthly trend chart

As Carlos **I want** a 12-month bar or area chart **So that** I can spot seasonal peaks

- **Priority:** Should · **Story points:** 2

Business rules

- Months Jan–Dec on X axis; tCO_2e on Y axis.
- Bars for months with no data show 0 (not missing).
- A horizontal dashed reference line shows the **average** across all months.
- Current month is highlighted with a subtle accent.

Acceptance criteria

1. **Given** I'm viewing 2026 and it's April **When** the chart renders **Then** Jan–Apr show data and May–Dec are 0 bars.

US-024 · Switch the dashboard year

As Emma **I want** to switch between 2021–2026 on the dashboard **So that** I can review historical performance

- **Priority:** Must · **Story points:** 2

Validations

- Year range: **2021–2026** (dynamic, based on configured years).

Business rules

- Year picker is a segmented control (MUI `ToggleButtonGroup`).
- 2021 is labelled "Baseline" with a green dot.
- 2026 is labelled "Current" with an amber dot (partial-year data).
- Changing year updates `?year=XXXX` in the URL and re-renders the server component.
- `useTransition` keeps the UI responsive during the navigation.

Acceptance criteria

1. **Given** I'm on `/dashboard?year=2024` **When** I click "2025" **Then** URL changes to `?year=2025` and all panels re-fetch without a hard reload.

US-025 · View top emitting activities

As Emma **I want** a table of the top 5 emitting activities for the year **So that** I know where to focus reduction efforts

- **Priority:** Should · **Story points:** 2

Business rules

- Table ranks activities by summed tCO_2e descending.
- Shows rank badge (#1–#5), activity name, scope chip, tCO_2e, and percentage share of total.
- Activity names are truncated with ellipsis if > 40 chars.

US-026 · View consumption breakdown

As Carlos **I want** to see total consumption per unit (kWh, litres, km, etc.) **So that** I understand activity volumes alongside their emissions

- **Priority:** Could · **Story points:** 2

Business rules

- Records are grouped by `unit`.
- For each unit: show total quantity and the top 3 contributing activities.
- Each unit group has a colour-coded chip for quick scanning.

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6. Epic 4 — Emissions Calculator (Scopes 1–3)

Goal: Let users log activity data and have the system compute CO_2e automatically.

US-030 · Log a Scope 1 emission record

As Carlos **I want** to log fuel consumption (natural gas, diesel, petrol, refrigerant leaks) **So that** our direct emissions are tracked

- **Priority:** Must · **Story points:** 5 · **Dependencies:** US-002

Validations

FIELD	RULE
Activity	Required · must belong to Scope 1 · must be `isActive = true`
Emission factor	Required · must belong to the selected activity
Quantity	Required · numeric · > 0 · max 15 decimal digits
Unit	Required · must match the activity's default unit (informational)
Period	Required · month/year picker · must be ≤ current month of current year
Notes	Optional · max 500 chars

Business rules

- CO_2e is computed server-side using `engine.calculate({ quantity, emissionFactorValue: factor.value })`.
- Formula: `co2eKg = quantity × factor.value`; `co2eT = co2eKg / 1000`.
- Results are rounded: `co2eKg` to 3 decimals, `co2eT` to 6 decimals.
- The `EmissionRecord` row stores both `co2eKg` and `co2eT` (denormalised for dashboard reads).
- The selected `emissionFactorId` is **immutable** once saved — changing the factor in the admin panel does NOT change historical records.
- On successful save, the API returns 201 with the full record (including joined activity + factor).

Acceptance criteria

1. **Given** I enter 12,400 kWh of natural gas with the UK DEFRA 2024 factor (0.18316 $kgCO_2e/kWh$) **When** I submit **Then** a record is created with `co2eKg = 2271.18` and `co2eT = 2.27118`.
2. **Given** I enter quantity `-5` **When** I submit **Then** I see "Quantity must be greater than 0" and no record is created.
3. **Given** I pick a period in the future (e.g. 2027-06) **When** I submit **Then** the API returns 400 with "Period cannot be in the future".
4. **Given** I select a deactivated activity **When** the form renders **Then** it doesn't appear in the dropdown.

Edge cases

- Quantity of 0 is rejected.
- `Infinity` or `NaN` from DevTools tampering is rejected by Zod.

US-031 · Log a Scope 2 emission record

As Carlos **I want** to log monthly grid electricity consumption with the right regional factor **So that** our purchased-energy emissions reflect the correct grid carbon intensity

- **Priority:** Must · **Story points:** 3

Business rules

- Same engine/validation as Scope 1.
- The factor picker defaults to the activity's `isDefault` factor but lets me pick alternatives (UK, US, EU).
- Scope 2 methodology is **location-based only** in Phase 1 (market-based RECs deferred).

Acceptance criteria

1. **Given** I log 48,000 kWh with UK DEFRA 2024 factor 0.20493 kgCO_2e/kWh **When** I submit **Then** `co2eKg = 9,836.64`.

US-032 · Log a Scope 3 emission record

As Carlos **I want** to log business travel, commuting, purchased goods, and waste **So that** our full value-chain footprint is captured

- **Priority:** Must · **Story points:** 3

Business rules

- Same engine/validation.
- Units are activity-specific (passenger-km, tonnes, kg).
- Notes are particularly useful here (e.g. flight origin/destination).

Acceptance criteria

1. **Given** I log 24,600 passenger-km of long-haul flight at 0.195 kgCO_2e/pkm **When** I submit **Then** `co2eKg = 4,797` and a new record appears in the All Records list.

US-033 · See factor details before saving

As Carlos **I want** to see the selected factor's source, region, and value before I commit **So that** I don't accidentally use the wrong one

- **Priority:** Should · **Story points:** 1

Acceptance criteria

1. **Given** I select a factor in the picker **When** the side panel updates **Then** I see "Source: DEFRA · Region: UK · Value: 0.20493 kgCO_2e/kWh · GWP: IPCC AR5 100-year".

US-034 · See a live CO_2e preview

As Carlos **I want** the form to show me the estimated CO_2e before I submit **So that** I can sanity-check the order of magnitude

- **Priority:** Could · **Story points:** 2

Acceptance criteria

1. **Given** I've selected an activity with factor 0.20493 and typed quantity 1000 **When** the form updates **Then** a "Preview" chip shows "≈ 204.93 kg CO_2e / 0.205 t CO_2e".

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7. Epic 5 — Emission Records Management

Goal: Let tenant users review, filter, and manage what's been logged.

US-040 · View all records

As Emma **I want** a paginated, filterable list of every emission record **So that** I can audit what my team has entered

- **Priority:** Must · **Story points:** 3

Validations

- Page size fixed at **20**; page number from `?page=` query param.

Business rules

- Columns: Date (period), Scope chip, Activity, Quantity + unit, CO_2e (t), Emission factor, User (who entered), Data source (MANUAL / CSV_IMPORT / API).
- Default sort: `period` descending.
- Scope chips are colour-coded consistently.

- Empty state shows an illustration and "No records yet — start with a Scope calculator".

Acceptance criteria

1. **Given** my entity has 450 records **When** I visit `/emissions` **Then** I see page 1 of 23 (20 per page), sorted newest first.
2. **Given** I click the Scope 2 chip on a row **When** the URL updates to `?scope=2` **Then** only Scope 2 records are shown.

US-041 · Filter records by year

As Emma **I want** to narrow records to a specific year **So that** I can produce period-specific audits

- **Priority:** Should · **Story points:** 1

Acceptance criteria

1. **Given** `/emissions` has a year dropdown **When** I pick "2025" **Then** URL changes to `?year=2025` and the list refreshes.

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8. Epic 6 — Reporting

Goal: Turn raw records into ISO 14064-aligned annual reports.

US-050 · Generate an annual report

As Emma **I want** to generate a GHG report for a specific year **So that** I can share aggregated data with auditors, investors, or regulators

- **Priority:** Must · **Story points:** 8 · **Dependencies:** US-030, US-031, US-032

Validations

FIELD	RULE
Year	Required · 2021 ≤ year ≤ current year · must have ≥ 1 record for that year · no existing report for that year

Business rules

- Report generation is triggered by `POST /api/reports` with `{ year }`.

- The API aggregates data from 4 parallel queries:
  1. All records for (entity, year) with scope + activity joined
  2. Previous-year aggregate for YoY computation
  3. 2021 baseline aggregate for vs-baseline computation
  4. Entity metadata (for the report title)
- The aggregated JSON **summary** is stored in `Report.summary` (JSONB).
- `Report.status` is set to `GENERATED`.
- If a report for the same (entity, year) already exists, API returns **409 Conflict**.
- If no records exist for the year, API returns **422 Unprocessable Entity** with a helpful message.

Summary JSON shape

```
{
  "totalCo2eT": 1248.5,
  "recordCount": 450,
  "scope1": { "co2eT": 342.1, "pct": 27.4, "activities": [] },
  "scope2": { "co2eT": 518.4, "pct": 41.5, "activities": [] },
  "scope3": { "co2eT": 388.0, "pct": 31.1, "activities": [] },
  "byMonth": [{ "month": 1, "co2eT": 92.3 }],
  "topActivities": [{ "name": "Grid electricity", "co2eT": 234.1, "scope"
  "baselineCo2eT": 1812.4,
  "prevYearCo2eT": 1418.0,
  "yoyChangePct": -11.95,
  "vsBaselinePct": -31.12,
  "entityName": "Acme Corp",
  "generatedAt": "2026-04-15T10:23:00Z"
}
```

Acceptance criteria

1. **Given** I click "Generate Report" and pick year 2025 **When** the API completes **Then** a new `Report` row is created with `status = GENERATED` and I see its detail panel inline.
2. **Given** a report for 2025 already exists **When** I try to generate another **Then** I see "A report for 2025 already exists — delete it first" and no new row is inserted.
3. **Given** 2027 has zero records **When** I try to generate a report for 2027 **Then** I see "No emission records found for 2027".

US-051 · View report detail panel

As Emma **I want** to view the full breakdown of a generated report inline **So that** I don't need to download a PDF just to review it

- **Priority:** Must · **Story points:** 5

Business rules

- Clicking a report row expands an inline panel showing:
  - 4 KPI cards (total emissions, YoY change, vs baseline, data quality)
  - Scope breakdown bars
  - 12-month bar chart
  - Top 5 emission sources with rank badges
- Data quality is derived from `recordCount`: High (≥50), Medium (20–49), Low (<20).

US-052 · Export report as CSV

As Carlos **I want** to download a report as CSV **So that** I can analyse it in Excel or feed it into another tool

- **Priority:** Must · **Story points:** 3

Business rules

- `GET /api/reports/[id]/export?format=csv` returns `text/csv` with `Content-Disposition: attachment`.
- Filename pattern: `carbonly-ghg-report-{year}.csv`.
- CSV contents (in this order):
  1. Report header block (organisation, year, title, generated timestamp, status)
  2. Summary block (total, prior year, YoY %, baseline, vs-baseline %)
  3. Scope breakdown (3 rows)
  4. Activities by scope (one row per activity)
  5. Monthly trend (12 rows: Jan–Dec)
  6. Top activities (rank, name, scope, tCO_2e)
  7. Footer credit line
- Values containing commas, quotes, or newlines are properly RFC 4180 escaped.

US-053 · Publish / unpublish a report

As Emma **I want** to mark a report as PUBLISHED (or revert to GENERATED) **So that** I can signal internally when a report is ready to share

- **Priority:** Should · **Story points:** 2

Business rules

- `PATCH /api/reports/[id]` with `{ status: "PUBLISHED" | "GENERATED" }`.
- Publish button only available for `GENERATED` reports.
- Unpublish button only available for `PUBLISHED` reports.

US-054 · Delete a report

As Emma **I want** to delete a mistakenly generated report **So that** I can re-generate with corrected data

- **Priority:** Should · **Story points:** 1

Business rules

- `DELETE /api/reports/[id]` removes the row.
- A `confirm()` dialog warns the action is irreversible.

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9. Epic 7 — Settings & Profile

Goal: Let users review their profile and organisation details.

US-060 · View my profile

As Carlos **I want** to see my name, email, role, and member-since date **So that** I can verify the account I'm signed into

- **Priority:** Should · **Story points:** 1

US-061 · View my organisation details

As Emma **I want** to see my organisation's name, slug, industry, country, and subscription status **So that** I can confirm the tenant context

- **Priority:** Should · **Story points:** 1

US-062 · View billing summary

As Emma **I want** a Billing section showing my current plan, status, and trial end date **So that** I know where my subscription stands

- **Priority:** Must · **Story points:** 3

Business rules

- Plan is resolved from `Entity.stripePriceId` via `getPlanByPriceId()`.
- Status chip uses colour mapping (ACTIVE=green, TRIALING=blue, PAST_DUE=amber, CANCELED=red).
- If `stripeCustomerId` is set, a "Manage billing" button opens the Stripe Customer Portal.
- If status is NONE / CANCELED / TRIALING, an "Upgrade plan" link routes to `/#pricing`.

Acceptance criteria

1. **Given** I'm on a TRIALING subscription ending on 2026-04-30 **When** I visit Settings **Then** I see "Trial ends in 15 days — 30 April 2026".

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10. Epic 8 — Billing & Subscriptions

Goal: Monetise via Stripe with three tiers and a 14-day trial.

US-070 · View pricing on the landing page

As Priya **I want** to see three plan tiers with features and prices **So that** I can evaluate Carbonly against alternatives

- **Priority:** Must · **Story points:** 3

Business rules

- Plans defined in `src/lib/stripe/plans.ts`:

PLAN	MONTHLY	ANNUAL	USERS	SCOPES	HISTORY	TRIAL
Starter	£49	£39	3	1, 2	1 year	14 days
Professional	£149	£119	15	1, 2, 3	unlimited	14 days
Enterprise	Custom	Custom	unlimited	1, 2, 3	unlimited	—

- Professional has a "Most Popular" badge and is visually highlighted.
- Enterprise CTA says "Contact sales" and opens a `mailto:sales@carbonly.io`.

US-071 · Start a paid plan via Stripe Checkout

As Emma **I want** to click "Start free trial" on a plan and go through Stripe Checkout **So that** my organisation gets onto that plan with a 14-day trial

- **Priority:** Must · **Story points:** 5

Validations

- User must be authenticated; if not, redirect to `/register`.
- User's `entityId` must exist.

Business rules

- `POST /api/stripe/checkout` with `{ planKey }` does:
  1. Ensure a `Customer` exists on Stripe; create if missing; save `stripeCustomerId`.
  2. Create a **Checkout Session** with `mode: "subscription"`, `trial_period_days: 14`, `success_url: /dashboard?upgraded=1`, `cancel_url: /pricing?canceled=1`.
  3. Return `{ url }` for client-side redirect.
- On success, Stripe fires `customer.subscription.created` → webhook → `Entity.subscriptionStatus = TRIALING`.

Acceptance criteria

1. **Given** I click "Start free trial" on Professional **When** the API responds **Then** I'm redirected to Stripe's hosted checkout with the correct price ID and trial config.
2. **Given** I cancel at Stripe **When** I return **Then** I land on `/pricing?canceled=1` and no subscription was created.

US-072 · Keep subscription state in sync via webhooks

As the system **I want** to mirror Stripe subscription state into the `Entity` table **So that** the UI reflects reality without polling

- **Priority:** Must · **Story points:** 5

Validations

- Webhook signature verified with `STRIPE_WEBHOOK_SECRET` via `stripe.webhooks.constructEvent()`.
- Unverified payloads return 400 without DB writes.

Business rules

- Events handled:

EVENT	ACTION
`customer.subscription.created`	Save `subscriptionStatus`, `stripeSubscriptionId`, `stripePriceId`
`customer.subscription.updated`	Same as created
`customer.subscription.deleted`	Set `subscriptionStatus = CANCELED`
`invoice.payment_failed`	Set `subscriptionStatus = PAST_DUE`

- Stripe status → Carbonly enum mapping:

STRIPE	CARBONLY
trialing	TRIALING
active	ACTIVE
past_due	PAST_DUE
canceled	CANCELED
unpaid	UNPAID
incomplete, incomplete_expired, paused	PAST_DUE (fallback: CANCELED)

US-073 · Open the Stripe Customer Portal

As Emma **I want** a "Manage billing" button that takes me to Stripe's hosted portal **So that** I can update my card, see invoices, and change plan

- **Priority:** Must · **Story points:** 2

Business rules

- `POST /api/stripe/portal` creates a **billing portal session** with `return_url: /settings`.
- Returns `{ url }` for client redirect.
- Only available when `stripeCustomerId` is set.

---

11. Epic 9 — Admin: Entity Management

Goal: Let platform admins manage tenant organisations.

US-080 · View all entities

As Adam **I want** a searchable list of every tenant organisation on the platform **So that** I can monitor and support them

- **Priority:** Must · **Story points:** 3

Business rules

- Table columns: Name, Slug (monospace), Industry, Country, Status chip, Users count, Records count, Created date.
- Stat cards above the table: Total, Active, Trialing, Churned (CANCELED + UNPAID).
- Search filters name or slug (client-side since dataset is small).

US-081 · Create a new entity

As Adam **I want** to create a new organisation via a sidesheet form **So that** a new customer is onboarded

- **Priority:** Must · **Story points:** 3

Validations

FIELD	RULE
Name	Required · 2–80 chars
Slug	Optional · if empty, auto-generated from name (lowercase, hyphen-separated, alphanumerics only) · must be unique
Industry	Optional · free text
Country	Optional · free text
Subscription status	Defaults to `TRIALING` · must be one of the enum values

Acceptance criteria

1. **Given** I open the "New Organisation" sidesheet and type "GreenTech Solutions" **When** I submit without a slug **Then** a row is created with slug `"greentech-solutions"`.
2. **Given** I submit with slug "acme-corp" which already exists **When** the form submits **Then** I see the error "Slug already in use".

US-082 · Edit an entity

As Adam **I want** to edit an entity's name, industry, country, or status **So that** I can keep records current

- **Priority:** Must · **Story points:** 2

US-083 · Delete an entity

As Adam **I want** to delete an entity and all its data **So that** I can offboard customers cleanly

- **Priority:** Should · **Story points:** 2

Business rules

- `DELETE /api/admin/entities/[id]` triggers Prisma cascade deletes: `UserEntity`, `EmissionRecord`, `Report`, `YearlyConfig`.
- A `confirm()` dialog warns: "This will remove all users, records and reports".

Acceptance criteria

1. **Given** an entity with 5 users, 450 records, 2 reports **When** I confirm deletion **Then** all associated rows are removed and the stat cards update.

---

12. Epic 10 — Admin: User Management

US-090 · View all users

As Adam **I want** a list of every registered user with their entity memberships **So that** I can audit access

- **Priority:** Must · **Story points:** 2

Business rules

- Columns: Name, Email, Role chip, Organisation, Joined.
- "You" badge on the current user row.
- Stat cards: Total, Super Admins, Admins, Experts.

US-091 · Invite (create) a user

As Adam **I want** to create a new user and assign them to an entity **So that** they can sign in immediately

- **Priority:** Must · **Story points:** 3

Validations

FIELD	RULE
Name	Optional · ≤ 80 chars
Email	Required · valid format · unique
Temporary password	Optional · if set, ≥ 8 chars
Global role	Required · EXPERT (default), ADMIN, or SUPER_ADMIN
Entity	Optional · if set, creates `UserEntity` row
Entity role	Required if entity is set · EXPERT (default) or ADMIN

Acceptance criteria

1. **Given** I invite `jane@acme-corp.com` as EXPERT at Acme Corp **When** I submit **Then** a user + entity membership row is created and she appears in the table.

US-092 · Edit a user

As Adam **I want** to edit a user's name or global role **So that** I can correct onboarding mistakes or promote users

- **Priority:** Must · **Story points:** 2

US-093 · Delete a user

As Adam **I want** to delete a user **So that** I can revoke access

- **Priority:** Must · **Story points:** 1

Business rules

- A user **cannot delete their own account** — API returns 400 "You cannot delete your own account".
- Cascade deletes: `UserEntity`, `Session`, `Account`.

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13. Epic 11 — Admin: Taxonomy Management

US-100 · View scopes & activities

As Adam **I want** to see each scope with its activities **So that** I can maintain the GHG Protocol taxonomy

- **Priority:** Must · **Story points:** 2

Business rules

- Three scope sections (Scope 1, 2, 3) with coloured headers and gradient icons.
- Each row: Activity name, description (truncated), Unit, Equation, Emission factor count, Active toggle.

US-101 · Add a new activity

As Adam **I want** to add a new activity under a scope **So that** I can keep up with evolving reporting requirements

- **Priority:** Must · **Story points:** 2

Validations

FIELD	RULE
Scope	Required · from existing scopes
Name	Required · ≤ 120 chars
Description	Optional · ≤ 500 chars
Unit	Required · ≤ 30 chars (e.g. "kWh", "litres", "passenger-km")
Equation	Optional · defaults to "quantity × emissionFactor"

Business rules

- Auto-increments `sortOrder` = max(sortOrder)+1 within the scope.
- Defaults to `isActive = true`.

US-102 · Edit / deactivate an activity

As Adam **I want** to update an activity or toggle its active state **So that** I can retire deprecated activities without breaking history

- **Priority:** Must · **Story points:** 2

Business rules

- Deactivated activities don't appear in the calculator dropdown but existing records are preserved.

US-103 · Delete an activity

As Adam **I want** to delete an activity that has no emission factors **So that** I can clean up truly unused entries

- **Priority:** Should · **Story points:** 1

Business rules

- `DELETE /api/admin/activities/[id]` returns **409 Conflict** if any `EmissionFactor` rows reference the activity.

---

14. Epic 12 — Admin: Emission Factor Management

US-110 · View all emission factors

As Adam **I want** a master table of every factor with its activity, value, unit, source, region, and default flag **So that** I can audit the taxonomy

- **Priority:** Must · **Story points:** 3

Business rules

- Columns: Scope chip, Activity, Factor name, Value (mono), Unit, Source, Region, Default chip.
- Search filters by factor name, activity name, or source.
- Scope filter dropdown (All / 1 / 2 / 3).
- Stats: Total factors, Defaults, per-scope counts.

US-111 · Add a new factor

As Adam **I want** to add a new factor to an existing activity **So that** users have the latest regulator-issued values

- **Priority:** Must · **Story points:** 3

Validations

FIELD	RULE
Activity	Required · must be active
Factor name	Required · ≤ 120 chars (e.g. "UK DEFRA 2024")
Value	Required · positive number · ≤ 15 decimal digits
Unit	Required · e.g. "kgCO2e/kWh"
Source	Optional
Region	Optional (e.g. "UK", "US", "EU")
GWP	Optional (e.g. "IPCC AR5 100-year")
Is default	Boolean

Business rules

- If `isDefault = true`, the API **automatically un-defaults all other factors for the same activity** (single-default rule).

Acceptance criteria

1. **Given** Activity "Grid electricity" has a default factor "DEFRA 2023" **When** I add a new factor "DEFRA 2024" marked as default **Then** DEFRA 2023 is automatically un-defaulted in the same transaction.

US-112 · Edit / toggle-default a factor

As Adam **I want** to update a factor's value or toggle its default flag **So that** I can roll over to a new year's factor without recreating records

- **Priority:** Must · **Story points:** 2

US-113 · Delete a factor

As Adam **I want** to delete a factor that isn't referenced by any records **So that** I can clean up unused entries

- **Priority:** Should · **Story points:** 1

Business rules

- Returns **409 Conflict** if `EmissionRecord` rows reference the factor.

15. Epic 13 — Admin: Platform Analytics & Billing Insights

US-120 · View platform overview

As Adam **I want** a dashboard of platform-wide KPIs **So that** I can monitor health at a glance

- **Priority:** Must · **Story points:** 5

Business rules

- Hero banner (dark gradient) with welcome text.
- KPI cards: Organisations, Users, Emission Records, CO_2e Tracked (tonnes).
- Subscription status breakdown (5-tile panel).
- Taxonomy summary panel (scopes, activities, factors, reports).
- Recent orgs list · Recent users list · Recent emission activity feed.

US-121 · View billing analytics

As Adam **I want** revenue-focused analytics (MRR, active customers, conversion, trials ending) **So that** I can track SaaS metrics

- **Priority:** Must · **Story points:** 5

Business rules

- MRR = sum of `monthlyPrice` across entities with `subscriptionStatus = ACTIVE` (priced via `getPlanByPriceId`).
- Conversion = % of all entities currently `ACTIVE`.
- Trials ending = entities with `subscriptionStatus = TRIALING` and `trialEndsAt ≤ now + 7 days`.
- Plan distribution bars (Starter / Professional / Enterprise / None).
- Subscription status cards (5 states).
- Alert banner showing trials ending soon with day countdown.
- Full subscription table with plan, status, trial end, stripe customer ID (truncated), users count.

Acceptance criteria

1. **Given** 3 active professional customers at £149/mo **When** I view `/admin/billing` **Then** MRR reads "£447".

---

16. Epic 14 — Marketing & Landing Page

US-130 · View the landing page

As Priya **I want** a polished landing page that explains Carbonly in 30 seconds **So that** I decide whether to sign up

- **Priority:** Must · **Story points:** 5

Business rules

- Sections: Hero (dark green gradient, typewriter headline) · Trusted-by industry chips · Animated stats bar with count-up · Features grid (6 items) · "How it works" with interactive 3-step auto-advance panel · Pricing · Testimonials · CTA banner · Footer.
- Landing nav is sticky; transitions to frosted glass on scroll.
- Responsive: single-column mobile, 2-col tablet, full layout desktop.
- All links to `/register` and `/login` work.

US-131 · Start free trial from pricing

As Priya **I want** "Start free trial" buttons on each plan card **So that** I can sign up directly from pricing

- **Priority:** Must · **Story points:** 2

Business rules

- Clicking calls `POST /api/stripe/checkout`; if unauthenticated (401), redirect to `/register`.
- Enterprise CTA opens `mailto:sales@carbonly.io?subject=Enterprise enquiry`.

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17. Cross-cutting acceptance criteria

These apply to **every** story:

17.1 Security

- ✅ No tenant data is ever returned or mutated without filtering by `entityId` from the current session.
- ✅ All destructive admin actions require `confirm()` on the client.
- ✅ Stripe webhooks are signature-verified before mutating state.
- ✅ No secret (Stripe key, NextAuth secret, DB password) is exposed to the client bundle.
- ✅ Server-sent HTML escapes all user-supplied strings (React defaults).

17.2 Error handling

- ✅ All API routes return JSON `{ error: string }` with an appropriate 4xx/5xx status.
- ✅ Client-side UI displays errors using MUI `Alert` components or inline helper text.
- ✅ Network failures show a user-friendly "Network error — please try again" message.
- ✅ 500s are logged server-side (Prisma error logs at `error` level in production).

17.3 Accessibility

- ✅ All interactive elements are keyboard accessible (Tab order, Enter/Space to activate).
- ✅ Form inputs have associated `<label>` elements.
- ✅ Colour is never the sole indicator of state (badges include text + icon).
- ✅ Focus rings are visible on all focusable elements.

17.4 Performance

- ✅ Initial page load ≤ 1.5s on a fast 3G connection.
- ✅ Dashboard data query ≤ 50ms on ~2,000 records.
- ✅ All pages use server components to avoid client waterfalls.

17.5 Responsive design

- ✅ All pages render correctly on 375px (mobile), 768px (tablet), 1280px (desktop).
- ✅ Admin tables support horizontal scroll on narrow viewports.

- ✅ Sidebar is a permanent drawer on ≥992px, temporary (hamburger) below.

17.6 Data integrity

- ✅ All foreign keys use cascading delete where ownership is clear (Entity → UserEntity/EmissionRecord/Report/YearlyConfig).
- ✅ Emission factor references in `EmissionRecord` are immutable post-creation.
- ✅ All CO_2e calculations are performed server-side; client inputs are never trusted.

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18. Traceability matrix

Each functional requirement in the BRD is covered by one or more user stories:

BRD REF	REQUIREMENT	COVERED BY
FR-AUTH-01	Email/password + Google auth	US-001, US-002, US-003
FR-AUTH-02	Session expiry	US-005
FR-AUTH-03	SUPER_ADMIN platform access	US-080 through US-121
FR-AUTH-04	Entity ADMIN manages their entity	US-061, US-062, US-073
FR-AUTH-05	EXPERT scoped to assigned entity	US-030, US-031, US-032
FR-AUTH-06	Onboarding redirect	US-010
FR-CALC-01	Scope 1/2/3 calculators	US-030, US-031, US-032
FR-CALC-02	Activity + factor selection	US-033
FR-CALC-03	`CO2e = quantity × factor`	US-030 (business rules)
FR-CALC-04	Store kg + tonnes	US-030 (business rules)
FR-CALC-05	Capture period + notes	US-030 validations
FR-CALC-06	Factor source/region/GWP	US-033, US-110
FR-DASH-01	Total by scope	US-020, US-021
FR-DASH-02	Donut chart	US-022
FR-DASH-03	Monthly trend	US-023
FR-DASH-04	Top 5 activities	US-025
FR-DASH-05	Year selection	US-024
FR-ENT-01	Entity CRUD	US-080, US-081, US-082, US-083
FR-ENT-02	Unique slug	US-081 validations
FR-ENT-03	Edit entity profile	US-082
FR-ENT-04	Stripe subscription mirror	US-072
FR-USR-01	View all users	US-090
FR-USR-02	Invite by email	US-091
FR-USR-03	Change role	US-092
FR-USR-04	Deactivation	US-093
FR-SCOPE-01	Activity CRUD	US-100, US-101, US-102, US-103

BRD REF	REQUIREMENT	COVERED BY
FR-SCOPE-02	Activity metadata	US-101 validations
FR-SCOPE-03	Factor CRUD	US-110, US-111, US-112, US-113
FR-SCOPE-04	Entity-specific overrides	Deferred (Phase 2)
FR-PAY-01	Three plans	US-070
FR-PAY-02	Stripe Checkout	US-071
FR-PAY-03	Webhook status sync	US-072
FR-PAY-04	Lapsed redirect to billing	US-062
FR-PAY-05	Stripe portal	US-073
FR-RPT-01	Generate annual report	US-050
FR-RPT-02	Scope + activity breakdown	US-051
FR-RPT-03	PDF + CSV export	US-052 (CSV); PDF deferred
FR-RPT-04	Status lifecycle	US-053
NFR-01	Tenant isolation	§17.1
NFR-02	RSC-first rendering	§17.4
NFR-05	Audit fields	US-030
NFR-06	Factor traceability	US-110, US-111